

Robinson George Arysseril

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EDUCATION

Vimal Jyothi Engineering College

B.Tech in Artificial Intelligence and Data Science CGPA: 9.46 / 10

- Currently pursuing Honors

Chemperi, Kerala

Sep. 2024 – May 2028

TECHNICAL SKILLS

Languages: C/C++, Rust, Java, Python, TypeScript, JavaScript, SQL, Xtensa Assembly

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Architecture, Memory Management

Tools & Frameworks: Git, GitHub Actions, Make, Linux, SDL2, Tauri, Angular, MySQL, REST APIs

EXPERIENCE

Research Intern

May 2026 – Jun. 2026

Amrita School of Artificial Intelligence

Coimbatore, India

- Designed and iterated on an NLP classification pipeline for bias detection under faculty mentorship, progressing through multiple modeling stages to reach 97.3% validation accuracy
- Incorporated feedback from research collaborators across iterations, refining model architecture and evaluation methodology to strengthen result reliability

Open Source Contributor & Maintainer

May 2023 – Present

Chultrabook Project

Remote

- Collaborate with a distributed team of maintainers across time zones to ship features, triage bug reports, and review contributions for a tool with 47,000+ users
- Work directly with end users to gather requirements, translating community feedback into prioritized engineering improvements
- Maintain release quality through regression testing across Windows, Linux, and macOS ahead of each version rollout

PROJECTS

Chultrabook Tools GitHub | Rust, Tauri, Angular, TypeScript

May 2023 – Present

- Built a cross-platform desktop application (Tauri, Rust, Angular) delivering real-time CPU temperature and fan-speed monitoring, custom fan curve profiles with boot-time persistence, RGB/keyboard backlight control, and keyboard remapping to 47,000+ users
- Architected a modular backend that decouples hardware-specific logic from the UI layer, enabling rapid support for new hardware generations with minimal code changes
- Packaged and maintained native distribution across Fedora/Terra, Debian/APT, and Arch/AUR alongside Windows and macOS installers, using GitHub Actions to automate cross-platform builds, growing the project to 128+ GitHub stars

NES Emulator GitHub | Rust, SDL2

Jul. 2026

- Implemented a cycle-accurate NES emulator with vsync-locked rendering, including a 6502 CPU decoded via bit-field opcode grouping that mirrors real hardware, a full PPU rendering pipeline, and support for 3 cartridge mappers (NROM, MMC1, CNROM)
- Traced and resolved layered bugs spanning CPU timing, bus address decoding, ROM parsing, and interrupt polling through systematic, hypothesis-driven debugging
- Verified correctness against Blargg's CPU instruction test suite and confirmed compatibility by running commercial ROMs end-to-end

Bare Metal ESP32 Kernel (null32) GitHub | C, Xtensa Assembly, Rust

Dec. 2025 – Present

- Built a bare-metal OS for the ESP32 (Xtensa LX6) from scratch: a two-stage bootloader, custom linker script, and NXE executable format with relocation patching to load and launch independently compiled user programs
- Wrote from-scratch UART and SPI drivers and a FAT32 filesystem with cluster-chain traversal and file read/write, exposed through an interactive on-device shell that loads and executes NXE binaries
- Designed a bitmap paged memory allocator tracking per-process page allocation via a process control block, and implemented the interrupt subsystem (custom vector table, timer/exception handlers) by working directly from the ESP32 technical reference manual

Terminal Movie Booking System GitHub | C, ncurses

Sep. 2025 – Oct. 2025

- Built a CLI cinema reservation system in C with an arrow-key-navigable ncurses UI and a from-scratch BMP parser that decodes raw pixel data to render movie posters as in-terminal ASCII art
- Modeled movies, showtimes, and customers as a nested linked-list hierarchy with a circular-queue booking pipeline, enforcing seat-availability and cancellation validation to prevent overbooking